

$$\begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ \vdots \end{bmatrix} \left[\begin{array}{c} + \\ + \\ \vdots \end{array} \begin{bmatrix} -1(0) + 0(0) + 1(0) \\ -2(0) + 0(1) + 2(2) \\ -1(0) + 0(1) + 1(1) \end{bmatrix} \begin{bmatrix} \dots \\ \dots \\ \ddots \end{bmatrix} \right]$$

The second matrix has a 3x3 submatrix highlighted in blue (rows 1-3, columns 1-3). Red arrows point from the center element (1) to its eight neighbors. The first row of the second matrix is $[0, 0, 0, 0, 0, 0]$. The first column is $[0, 0, 0, 1, 0, 0]^T$. The second column is $[0, 1, 1, 0, 0, 1]^T$. The third column is $[0, 2, 1, 0, 1, 1]^T$. The fourth column is $[0, 1, 1, 0, 1, 1]^T$. The fifth column is $[0, 1, 1, 0, 1, 1]^T$. The sixth column is $[0, 2, 1, 0, 0, 1]^T$.